

实验报告:

实验名称: 氢原子光谱

一. 数据处理.

1. 计算光栅常数 d .

测量次数	测量级次	-2级	+2级			
		d_1	β_1	d_2	β_2	$2\theta = \frac{1}{2}(d_1 - d_2 + \beta_1 - \beta_2)$
1		$30^\circ 01'$	$12^\circ 01'$	$267^\circ 25'$	$87^\circ 31'$	$41^\circ 28'$
2		$25^\circ 50'$	$205^\circ 49'$	$344^\circ 30'$	$164^\circ 31'$	$41^\circ 19'$
3		$75^\circ 42'$	$255^\circ 40'$	$34^\circ 20'$	$214^\circ 18'$	$41^\circ 22'$
4		$88^\circ 01'$	$263^\circ 00'$	$4^\circ 46'$	$221^\circ 46'$	$41^\circ 14.5'$
5		$19^\circ 08'$	$11^\circ 08'$	$149^\circ 45'$	$329^\circ 46'$	$41^\circ 22.5'$

$$\bar{\theta} = \frac{1}{5} \sum_{i=1}^5 \theta_i = \frac{1}{5} (20.7333^\circ + 20.6583^\circ + 20.6833^\circ + 20.6208^\circ + 20.6875^\circ) = 20.6766^\circ$$

$$\text{由 } d \sin \theta = 2\lambda \text{ 得 光栅常数 } d = \frac{2\lambda}{\sin \theta} = \frac{2 \times 589.3 \text{ nm}}{\sin(20.6766^\circ)} = 3.33793 \times 10^{-6} \text{ m}$$

计算不确定度:

$$U_A(\theta) = \sqrt{\frac{\sum_{i=1}^5 (\theta_i - \bar{\theta})^2}{5 \times 4}} = \sqrt{\frac{(20.7333 - 20.6766)^2 + (20.6583 - 20.6766)^2 + (20.6833 - 20.6766)^2 + (20.6208 - 20.6766)^2 + (20.6875 - 20.6766)^2}{20}}$$

$$= 0.0184758 = 3.224635 \times 10^{-4} \text{ rad.}$$

$$U_B(\theta) = \frac{0.5'}{\sqrt{3}} = 8.31722 \times 10^{-5} \text{ rad.}$$

$$U(\theta) = \sqrt{U_A^2(\theta) + U_B^2(\theta)} = 3.330076 \times 10^{-4} \text{ rad}$$

$$U(d) = d \cdot \sqrt{\left[\frac{U(\theta)}{\tan \theta} \right]^2} = \sqrt{\left(\frac{3.330076 \times 10^{-4}}{\tan(20.6766^\circ)} \right)^2 \times 3.33793 \times 10^{-6}} = 2.14529 \times 10^{-9} \text{ m}$$

∴ 最终结果: $d \pm U(d) = (3.338 \pm 0.003) \times 10^{-6} \text{ m}$

2. 计算氢原子里德伯常数 R_H .

级次	-1		+1		-2		+2	
颜色	α_1	β_1	α_1	β_1	α_2	β_2	α_2	β_2
红	$182^\circ 55'$	$2^\circ 55'$	$160^\circ 10'$	$340^\circ 08'$	$194^\circ 30'$	$14^\circ 30'$	$148^\circ 25'$	$328^\circ 20'$
蓝	$179^\circ 50'$	$359^\circ 50'$	$163^\circ 30'$	$343^\circ 30'$	$188^\circ 26'$	$8^\circ 24'$	$154^\circ 11'$	$334^\circ 12'$
紫	$179^\circ 00'$	$359^\circ 00'$	$163^\circ 59'$	$344^\circ 00'$	$186^\circ 28'$	$6^\circ 30'$	$156^\circ 08'$	$336^\circ 10'$

	-3		+3	
	α_3	β_3	α_3	β_3
红	$206^\circ 18'$	$26^\circ 18'$	$133^\circ 52'$	$313^\circ 50'$
蓝	$197^\circ 10'$	$17^\circ 55'$	$145^\circ 20'$	$325^\circ 10'$
紫	$194^\circ 20'$	$14^\circ 18'$	$148^\circ 00'$	$328^\circ 00'$

红光: ± 1 级: $\theta_1 = 11^\circ 23'$ ± 2 级: $\theta_2 = 23^\circ 03.75'$ ± 3 级: $\theta_3 = 36^\circ 13.5'$

蓝光: ± 1 级: $\theta_1 = 8^\circ 10'$ ± 2 级: $\theta_2 = 17^\circ 06.75'$ ± 3 级: $\theta_3 = 26^\circ$

紫光: ± 1 级: $\theta_1 = 7^\circ 30.25'$ ± 2 级: $\theta_2 = 15^\circ 10'$ ± 3 级: $\theta_3 = 23^\circ 9.5'$

整理得: 红 $\theta_1 = 11.3833^\circ$ $\theta_2 = 23.0625^\circ$ $\theta_3 = 36.225^\circ$

蓝 $\theta_1 = 8.1667^\circ$ $\theta_2 = 17.1125^\circ$ $\theta_3 = 26^\circ$

紫 $\theta_1 = 7.5042^\circ$ $\theta_2 = 15.1667^\circ$ $\theta_3 = 23.1583^\circ$

$$\text{红光: } \lambda_1 = \frac{d \sin \theta_1}{1} = 6.58826 \times 10^{-7} \text{ m} \quad \mu(\lambda_1) = \mu(d) \sin \theta_1 + d \cos \theta_1 \cdot \mu(\theta_1) = 8.66894 \times 10^{-10} \text{ m}$$

$$\lambda_2 = \frac{d \sin \theta_2}{2} = 6.53806 \times 10^{-7} \text{ m} \quad \mu(\lambda_2) = \frac{1}{2} [\mu(d) \sin \theta_2 + d \cos \theta_2 \mu(\theta_2)] = 7.16548 \times 10^{-10} \text{ m}$$

$$\lambda_3 = \frac{d \sin \theta_3}{3} = 6.57539 \times 10^{-7} \text{ m} \quad \mu(\lambda_3) = \frac{1}{3} [\mu(d) \sin \theta_3 + d \cos \theta_3 \mu(\theta_3)] = 6.66329 \times 10^{-10} \text{ m}$$

$$\bar{\lambda}_{\text{红}} = \left[\frac{\lambda_1}{\mu^2(\lambda_1)} + \frac{\lambda_2}{\mu^2(\lambda_2)} + \frac{\lambda_3}{\mu^2(\lambda_3)} \right] / \left[\frac{1}{\mu^2(\lambda_1)} + \frac{1}{\mu^2(\lambda_2)} + \frac{1}{\mu^2(\lambda_3)} \right] = 6.56534 \times 10^{-7} \text{ m}$$

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$$u(\bar{\lambda}_{\text{红}}) = \sqrt{1 / \left[\frac{1}{u^2(\lambda_1)} + \frac{1}{u^2(\lambda_2)} + \frac{1}{u^2(\lambda_3)} \right]} = 4.25221 \times 10^{-10} \text{ m}$$

$$R_H(\text{红}) = \frac{1}{\bar{\lambda}_{\text{红}}} \cdot \frac{1}{\frac{1}{\lambda_2^2} - \frac{1}{\lambda_1^2}} = 1.09667 \times 10^7 \text{ m}^{-1}$$

$$u(R_H(\text{红})) = \frac{1}{\frac{1}{\lambda_2^2} - \frac{1}{\lambda_1^2}} \cdot \frac{1}{\bar{\lambda}_{\text{红}}^2} \cdot u(\bar{\lambda}_{\text{红}}) = 7.10285 \times 10^3 \text{ m}^{-1}$$

$$\therefore R_H(\text{红}) \pm u(R_H(\text{红})) = (1.09667 \pm 0.0007) \times 10^7 \text{ m}^{-1}$$

$$\text{蓝光: } \lambda_1 = \frac{d \sin \theta_1}{1} = 4.74165 \times 10^{-7} \text{ m} \quad u(\lambda_1) = u(d) \sin \theta_1 + d \cos \theta_1 u(\theta_1) = 7.08612 \times 10^{-10} \text{ m}$$

$$\lambda_2 = \frac{d \sin \theta_2}{2} = 4.91091 \times 10^{-7} \text{ m} \quad u(\lambda_2) = \frac{1}{2} [u(d) \sin \theta_2 + d \cos \theta_2 u(\theta_2)] = 5.75315 \times 10^{-10} \text{ m}$$

$$\lambda_3 = \frac{d \sin \theta_3}{3} = 4.87751 \times 10^{-7} \text{ m} \quad u(\lambda_3) = \frac{1}{3} [u(d) \sin \theta_3 + d \cos \theta_3 u(\theta_3)] = 5.22346 \times 10^{-10} \text{ m}$$

$$\bar{\lambda}_{\text{蓝}} = \left[\frac{\lambda_1}{u^2(\lambda_1)} + \frac{\lambda_2}{u^2(\lambda_2)} + \frac{\lambda_3}{u^2(\lambda_3)} \right] / \left[\frac{1}{u^2(\lambda_1)} + \frac{1}{u^2(\lambda_2)} + \frac{1}{u^2(\lambda_3)} \right] = 4.85759 \times 10^{-7} \text{ m}$$

$$u(\bar{\lambda}_{\text{蓝}}) = \sqrt{1 / \left[\frac{1}{u^2(\lambda_1)} + \frac{1}{u^2(\lambda_2)} + \frac{1}{u^2(\lambda_3)} \right]} = 3.38910 \times 10^{-10} \text{ m}$$

$$R_H(\text{蓝}) = \frac{1}{\bar{\lambda}_{\text{蓝}}} \cdot \frac{1}{\frac{1}{\lambda_2^2} - \frac{1}{\lambda_1^2}} = 1.09793 \times 10^7 \text{ m}^{-1}$$

$$u(R_H(\text{蓝})) = \frac{1}{\frac{1}{\lambda_2^2} - \frac{1}{\lambda_1^2}} \cdot \frac{1}{\bar{\lambda}_{\text{蓝}}^2} \cdot u(\bar{\lambda}_{\text{蓝}}) = 7.66022 \times 10^3 \text{ m}^{-1}$$

$$\therefore R_H(\text{蓝}) \pm u(R_H(\text{蓝})) = (1.09793 \pm 0.0007) \times 10^7 \text{ m}^{-1}$$

$$\text{紫光: } \lambda_1 = \frac{d \sin \theta_1}{1} = 4.35930 \times 10^{-7} \text{ m} \quad u(\lambda_1) = u(d) \sin \theta_1 + d \cos \theta_1 u(\theta_1) = 6.69689 \times 10^{-10} \text{ m}$$

$$\lambda_2 = \frac{d \sin \theta_2}{2} = 4.36648 \times 10^{-7} \text{ m} \quad u(\lambda_2) = \frac{1}{2} [u(d) \sin \theta_2 + d \cos \theta_2 u(\theta_2)] = 5.27728 \times 10^{-10} \text{ m}$$

$$\lambda_3 = \frac{d \sin \theta_3}{3} = 4.37572 \times 10^{-7} \text{ m} \quad u(\lambda_3) = \frac{1}{3} [u(d) \sin \theta_3 + d \cos \theta_3 u(\theta_3)] = 4.79175 \times 10^{-10} \text{ m}$$

$$\bar{\lambda}_{\text{紫}} = \left[\frac{\lambda_1}{u^2(\lambda_1)} + \frac{\lambda_2}{u^2(\lambda_2)} + \frac{\lambda_3}{u^2(\lambda_3)} \right] / \left[\frac{1}{u^2(\lambda_1)} + \frac{1}{u^2(\lambda_2)} + \frac{1}{u^2(\lambda_3)} \right] = 4.36886 \times 10^{-7} \text{ m}$$

$$u(\bar{\lambda}_{\text{紫}}) = \sqrt{1 / \left[\frac{1}{u^2(\lambda_1)} + \frac{1}{u^2(\lambda_2)} + \frac{1}{u^2(\lambda_3)} \right]} = 3.13482 \times 10^{-10} \text{ m}$$

$$R_H(\text{紫}) = \frac{1}{\bar{\lambda}_{\text{紫}}} \cdot \frac{1}{\frac{1}{\lambda_2^2} - \frac{1}{\lambda_1^2}} = 1.08997 \times 10^7 \text{ m}^{-1}$$

$$u(R_H(\text{紫})) = \frac{1}{\frac{1}{\lambda_2^2} - \frac{1}{\lambda_1^2}} \cdot \frac{1}{\bar{\lambda}_{\text{紫}}^2} \cdot u(\bar{\lambda}_{\text{紫}}) = 7.82091 \times 10^3 \text{ m}^{-1}$$

$$\therefore R_H(\text{紫}) \pm u(R_H(\text{紫})) = (1.08997 \pm 0.0008) \times 10^7 \text{ m}^{-1}$$

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$$\text{总体: } \bar{R}_H = \left[\frac{R_H(\text{红})}{n^2(R_H(\text{红}))} + \frac{R_H(\text{蓝})}{n^2(R_H(\text{蓝}))} + \frac{R_H(\text{紫})}{n^2(R_H(\text{紫}))} \right] / \left[\frac{1}{n^2(R_H(\text{红}))} + \frac{1}{n^2(R_H(\text{蓝}))} + \frac{1}{n^2(R_H(\text{紫}))} \right]$$

$$= 1.04500 \times 10^7 \text{ m}^{-1}$$

$$u(\bar{R}_H) = \sqrt{1 / \left[\frac{1}{n^2(R_H(\text{红}))} + \frac{1}{n^2(R_H(\text{蓝}))} + \frac{1}{n^2(R_H(\text{紫}))} \right]} = 4.2082 \times 10^3 \text{ m}^{-1}$$

$$\therefore \bar{R}_H \pm u(\bar{R}_H) = (1.0450 \pm 0.0004) \times 10^7 \text{ m}^{-1}$$

$$\text{相对误差 } \eta = \left| \frac{\bar{R}_H - R_{H\text{标}}}{R_{H\text{标}}} \right| \times 100\% = \left| \frac{1.0450 - 1.047373}{1.047373} \right| \times 100\% = 0.21\%$$

3. 计算钠黄光 $k = \pm 2$ 级谱线的角色散率和分辨本领

① 角色散率 $D_\theta = \frac{k}{d \cos \theta}$

当 $k=2$ 时 $D_{\theta 2} = \frac{2}{d \cos \theta} = \frac{2}{2.33793 \times 10^{-6} \cdot \cos(20.6766^\circ)} = 6.40425 \times 10^5 \text{ rad/m}$

② 色分辨本领 $R = \frac{\lambda}{\Delta \lambda} = kN$

$$N = \frac{D}{d} = \frac{2.20 \text{ cm}}{2.33793 \times 10^{-6} \text{ m}} = \frac{2.2}{2.33793} \times 10^6 = 6591 \text{ 条}$$

当 $k=2$ 时 $R_2 = kN = 13182 \text{ 条}$

三 课后思考题

1. 钠黄光 $\lambda = 589.3 \text{ nm}$ 其中双线的波长分别为 589 nm 和 589.6 nm

$$\therefore \Delta \lambda = 0.6 \text{ nm}$$

$$\frac{\lambda}{\Delta \lambda} = \frac{589.3}{0.6} = 982.17 < 6591 < 13182$$

由此可知 本实验装置可以将钠黄光的双线分开。

利用 1.2 级衍射光即可将钠黄光的双线分开。

2. ① 化学分析: 利用光谱技术测定物质分子结构与化学成分。

② 环境监测: 对环境样品监测水质、空气污染。

③ 天文学: 通过天体发出的光线了解其状态、成分。

第一次互查: (望远镜, 载物台). 杨鹏飞.

第二次互查: (光栅). 杨鹏飞.

-2

+2

	α_1	β_1	α_2	β_2	$2\theta = \frac{1}{2}(\alpha_1 - \alpha_2 + \beta_1 - \beta_2)$
1	$30^\circ 01'$	$12^\circ 01'$	$267^\circ 35' 41^{26}$	$87^\circ 31' 41^{30}$	$41^\circ 28' \checkmark$
2	$25^\circ 50'$	$205^\circ 49'$	$344^\circ 30' 41^{20}$	$164^\circ 31' 41^{18}$	$41^\circ 19' \checkmark$
3	$75^\circ 42'$	$255^\circ 40'$	$34^\circ 20' 41^{22}$	$214^\circ 18' 41^{22}$	$41^\circ 22' \checkmark$
4	$83^\circ 01'$	$263^\circ 00'$	$41^\circ 46' 41^{15}$	$221^\circ 46' 41^{14}$	$41^\circ 145' \checkmark$
5	$191^\circ 08'$	$11^\circ 08'$	$149^\circ 45' 41^{23}$	$329^\circ 46' 41^{22}$	$41^\circ 225' \checkmark$

对三片

红 $2\theta_1 - 22^\circ 46' \checkmark$ $2\theta_2 46^\circ 075' \checkmark$ $2\theta_3 72^\circ 27' \checkmark$

蓝 $2\theta_1 16^\circ 20' \checkmark$ $2\theta_2 34^\circ 135' \checkmark$ $2\theta_3 34^\circ 12' \checkmark$

紫 $2\theta_1 - 115^\circ 005' \checkmark +1 30^\circ 20' - 2 \checkmark$ $2\theta_3 + 2 46^\circ 19' \checkmark$

对三片

	α_1	β_1	α_1	β_1	α_2	β_2	α_2	β_2
红	$182^\circ 55'$	$2^\circ 55'$			$194^\circ 30'$	$14^\circ 30'$		
蓝	$179^\circ 50'$	$359^\circ 50'$			$188^\circ 26'$	$8^\circ 24'$		
紫	$179^\circ 00'$	$259^\circ 00'$	$160^\circ 10'$	$340^\circ 08'$	$186^\circ 28'$	$6^\circ 30'$	$148^\circ 25'$	$328^\circ 20'$
			$163^\circ 30'$	$343^\circ 30'$			$154^\circ 11'$	$334^\circ 12'$
			$163^\circ 59'$	$344^\circ 00'$			$156^\circ 08'$	$336^\circ 10'$

-3

+3

α_3	β_3	α_3	β_3
$206^\circ 18'$	$26^\circ 18'$	$133^\circ 52'$	$313^\circ 50'$
$197^\circ 10'$	$17^\circ 05'$	$145^\circ 20'$	$325^\circ 10'$
$194^\circ 20'$	$14^\circ 18'$	$148^\circ 00'$	$328^\circ 00'$

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